

Edge-AI-Based Adaptive Traffic Signal Control for Emergency Vehicle Priority

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Abstract—Urban traffic congestion can increase vehicle waiting time and make it difficult for emergency vehicles such as ambulances to move efficiently through busy intersections. Conventional traffic signals generally operate using fixed timing plans and therefore have limited ability to respond to rapidly changing traffic conditions. Recent research has explored adaptive traffic signal control using machine vision, artificial intelligence, and reinforcement learning. This paper proposes an edge-AI-based adaptive traffic signal framework that uses roadside cameras and local computing units to analyze traffic conditions and provide traffic-signal decisions with reduced dependence on continuous cloud communication. The proposed framework additionally focuses on emergency vehicle priority, where an ambulance can be detected and nearby traffic signals can be coordinated to provide a clearer route. The architecture combines roadside sensing, local processing, optional cloud-based storage and analysis, and traffic-signal control. The proposed approach is intended to provide a practical foundation for developing a responsive traffic management system suitable for intelligent transportation applications.

Index Terms—Edge AI, adaptive traffic signal control, intelligent transportation system, machine vision, emergency vehicle priority, traffic management.

I. INTRODUCTION

Traffic congestion is a major challenge in urban transportation systems. Traditional traffic signals commonly operate using predefined timing schedules. Although fixed-time control is simple to implement, it may not respond effectively when traffic density changes unexpectedly.

Artificial intelligence and machine vision provide an opportunity to make traffic signal systems more responsive. Camera-based systems can collect information about vehicles and traffic queues, while intelligent algorithms can use this information to modify signal timing according to current traffic conditions.

Research in traffic signal control has investigated machine learning and deep reinforcement learning approaches for adapting traffic signals to changing traffic conditions [1]. Machine-vision-based traffic monitoring has also been studied for estimating traffic-flow parameters and adapting traffic-light operation [2].

However, an important practical requirement is the handling of emergency vehicles. An ambulance travelling through congested intersections may experience delays if traffic signals continue to operate according to normal timing schedules. Therefore, there is a need for a traffic signal framework

that can combine normal adaptive traffic management with emergency vehicle priority.

This paper presents a proposed edge-AI-based architecture for adaptive traffic signal control with a specific focus on emergency vehicle priority.

II. LITERATURE REVIEW

A. AI-Based Traffic Signal Control

Artificial intelligence has been investigated as an approach for improving traffic signal control. Rasheed *et al.* reviewed deep reinforcement learning techniques for traffic signal control and discussed their application to traffic congestion and adaptive signal management [1].

The use of learning-based methods allows traffic signal systems to consider traffic conditions rather than relying only on predetermined schedules. However, practical deployment requires suitable sensing, computing resources, communication infrastructure, and reliable control mechanisms.

B. Machine Vision for Adaptive Traffic Signals

Machine vision provides a way to obtain traffic information directly from road-side cameras. Gorodokin *et al.* investigated the use of machine vision and neural-network-based vehicle tracking for adaptive traffic-light control [2].

Their work demonstrates the potential of using parameters such as vehicle counts and queue conditions to adjust traffic signal operation dynamically.

C. Edge Processing

Cloud-based artificial intelligence systems can provide significant computational resources, but transmitting continuous camera information to a remote server can increase communication requirements and introduce network-dependent delays.

An edge-computing approach moves at least part of the processing closer to the traffic intersection. In the proposed framework, a local computing unit receives camera information, performs traffic analysis, and generates a signal-control decision. Only selected information may be forwarded to a cloud server for storage or wider analysis.

III. PROBLEM STATEMENT

Conventional fixed-time traffic signals are unable to respond effectively to sudden changes in traffic density. Although adaptive traffic signal systems can use traffic information

to modify signal operation, emergency vehicles may still encounter delays when normal traffic-control logic is followed.

The problem addressed in this work is therefore:

To design a traffic signal control framework that can analyze local traffic conditions using edge-based artificial intelligence and provide priority to detected emergency vehicles while maintaining normal adaptive traffic management.

IV. RESEARCH GAP

Existing research has investigated adaptive traffic signal control using machine vision and artificial intelligence. However, there remains a practical requirement for systems that integrate local traffic analysis with emergency vehicle priority.

The research gap considered in this paper is the integration of:

- local camera-based traffic monitoring,
- edge-based traffic analysis,
- adaptive signal control, and
- emergency vehicle priority.

The proposed framework addresses this gap at the system-architecture level.

V. PROPOSED SYSTEM

The proposed system consists of four major stages:

- 1) Street cameras capture the traffic conditions at an intersection.
- 2) A local edge-computing unit processes the camera information and estimates the current traffic condition.
- 3) The system checks for the presence of an emergency vehicle and determines whether priority control is required.
- 4) The traffic-light controller modifies the signal sequence according to the generated decision.

Selected traffic information can optionally be transmitted to a cloud server for storage, monitoring, and large-scale analysis.

The overall system can be represented as:

Camera → Edge AI → Decision Module → Traffic Signal (1)

An optional cloud connection is represented as:

Edge AI → Cloud Server (2)

VI. SYSTEM ARCHITECTURE

The proposed architecture contains the following components.

A. Street Cameras

Cameras installed near the intersection continuously capture information about vehicles and road conditions. The camera input provides the primary source of traffic information for the system.

B. Edge Computing Unit

The edge computing unit processes the camera input locally. It can perform vehicle detection, traffic-density estimation, queue estimation, and emergency-vehicle detection.

Local processing reduces the requirement to continuously transmit raw video to a remote server.

C. Emergency Vehicle Detection

The emergency vehicle detection module identifies an ambulance or other authorized emergency vehicle from the available traffic information.

When an emergency vehicle is detected, the system generates an emergency-priority request.

D. Decision Module

The decision module combines traffic information with emergency-priority information.

Under normal conditions, the system can select signal timing based on traffic density and queue conditions. When an emergency vehicle is detected, the system can initiate a priority sequence for the required direction.

E. Traffic Signal Controller

The controller receives the decision from the local processing unit and changes the traffic signal accordingly.

The controller must also maintain appropriate safety constraints so that conflicting traffic movements are not simultaneously given a green signal.

F. Cloud Server

The cloud server is an optional component. It can be used for storing selected traffic information, monitoring system performance, and conducting wider traffic analysis.

The proposed architecture does not require continuous transmission of raw camera data to the cloud for every traffic decision.

VII. WORKING PRINCIPLE

The proposed system operates through the following sequence:

- 1) The camera captures the current road scene.
- 2) The edge computing unit receives the camera input.
- 3) The AI-based vision module detects vehicles and estimates traffic conditions.
- 4) The system checks whether an emergency vehicle is present.
- 5) If no emergency vehicle is detected, signal timing is adjusted according to the current traffic condition.
- 6) If an emergency vehicle is detected, the system generates an emergency-priority decision.
- 7) The traffic signal controller changes the appropriate signal sequence to support the movement of the emergency vehicle.
- 8) After the emergency vehicle passes the controlled intersection, the system returns to normal adaptive operation.

VIII. COMPARISON OF TRAFFIC SIGNAL APPROACHES

A conceptual comparison of the three approaches is presented in Table I.

TABLE I
CONCEPTUAL COMPARISON OF TRAFFIC SIGNAL APPROACHES

Approach	Decision Method	Internet Dependence	Adaptability
Fixed-Time Signal	Predefined timer	Very Low	Low
Cloud-Based AI	Remote AI processing	High	High
Edge-AI-Based System	Local AI processing	Low/Optional	High

The comparison is conceptual and is intended to describe the architectural differences. Actual response time, cost, and communication requirements depend on the hardware, software, network, and deployment environment.

IX. ADVANTAGES

The proposed framework provides the following potential advantages:

- Local processing of traffic information.
- Reduced dependence on continuous cloud communication.
- Adaptive response to changing traffic conditions.
- Support for emergency vehicle priority.
- Possibility of storing selected traffic information for future analysis.
- Scalable architecture for intelligent transportation applications.

X. LIMITATIONS

The proposed framework also has several limitations.

First, reliable traffic analysis depends on camera quality, lighting conditions, weather, camera placement, and the performance of the AI model.

Second, emergency vehicle detection must be sufficiently reliable because an incorrect detection could result in an unnecessary signal-priority request.

Third, the traffic-signal controller must enforce safety constraints before implementing any AI-generated decision.

Finally, the proposed architecture has not been presented as a field-tested system in this paper. Quantitative claims about response time, accuracy, cost, or reduction in ambulance travel time require experimental validation.

XI. FUTURE SCOPE

Future work can focus on implementing the proposed architecture using an edge computing platform and a suitable computer-vision model.

The system can be evaluated using traffic simulations and controlled real-world experiments. Important evaluation parameters can include:

- emergency vehicle detection accuracy,
- vehicle detection accuracy,

- average vehicle waiting time,
- emergency vehicle clearance time,
- signal decision latency,
- network data usage, and
- system reliability under different traffic conditions.

Future research can also investigate multi-intersection coordination, where several traffic signals cooperate to create a continuous emergency vehicle route.

XII. CONCLUSION

This paper proposed an edge-AI-based adaptive traffic signal framework for emergency vehicle priority. The proposed system combines camera-based traffic monitoring, local AI processing, adaptive traffic signal control, and optional cloud-based analysis.

Existing research demonstrates the potential of artificial intelligence, machine vision, and adaptive traffic control for responding to changing traffic conditions [1], [2]. Building on these approaches, the proposed framework focuses specifically on integrating emergency vehicle priority with local traffic processing.

The framework provides a basis for developing and evaluating an intelligent traffic management system. However, its effectiveness must be established through implementation and experimental evaluation rather than assumed from the proposed architecture.

REFERENCES

- [1] F. Rasheed, K.-L. A. Yau, R. M. Noor, C. Wu, and Y.-C. Low, "Deep Reinforcement Learning for Traffic Signal Control: A Review," *IEEE Access*, vol. 8, pp. 208016–208044, 2020, doi: 10.1109/ACCESS.2020.3034141.
- [2] V. Gorodokin, S. Zhankaziev, E. Shepeleva, K. Magdin, and S. Evtyukov, "Optimization of adaptive traffic light control modes based on machine vision," *Transportation Research Procedia*, vol. 57, pp. 241–249, 2021, doi: 10.1016/j.trpro.2021.09.047.